```
void printTime() {
  char buffer[3];
  const char* AMPM = 0;

  readTime();
  Serial.print(days[weekday-1]);
  Serial.print(" ");
  Serial.print(months[month-1]);
  Serial.print(" ");
  Serial.print(monthday);
  Serial.print(", 20");
  Serial.print(year);
  Serial.print(" ");

  if (hour > 12) {
    hour -= 12;
    AMPM = " PM";
  } else AMPM = " AM";

  Serial.print(hour);
  Serial.print(":");

  sprintf(buffer, "%02d", minute);
  Serial.print(buffer);
  Serial.println(AMPM);
}

void readTime() {
  Wire.beginTransmission(DS1307);
  Wire.write(byte(0));
  Wire.endTransmission();

  Wire.requestFrom(DS1307, 7);

  second = bcdToDec(Wire.read());
  minute = bcdToDec(Wire.read());
  hour = bcdToDec(Wire.read());
  weekday = bcdToDec(Wire.read());
  monthday = bcdToDec(Wire.read());
  month = bcdToDec(Wire.read());
  year = bcdToDec(Wire.read());
}
```

Whew. Yeah this one is a lot longer. Although most of this sketch is just repeated from the previous version in Listing 10-3 and the rest of it is taken up in setting the date and time and providing an interface so that we can enter the details. With this sketch uploaded and the Serial Monitor opened, we should be presented with something similar to the following text:

```
The current date and time is: Thursday July 7, 2011 9:53 PM
To set the date and time please select Newline ending to continue.
Would you like to set the date and time now? Y/N
```